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Superparamagnetism

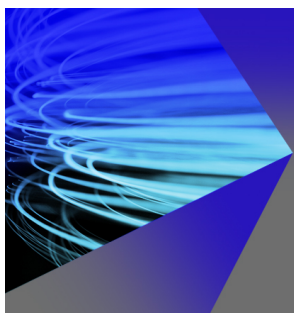
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Superparamagnetism

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Single domain ferromagnetic particles at elevated temperatures can behave magnetically in a manner analogous to the Langevin paramagnetism of moment-bearing atoms. The main distinction is that the moment of the particle may be 10^4 times the atomic moment. The experimental and theoretical foundations of this effect are reviewed and indication is made of the applications of this phenomenon. Lastly, attention is drawn to some current problems in this area.

THERMAL EQUILIBRIUM PROPERTIES OF SINGLE DOMAIN PARTICLES

FRENKEL and Dorfman¹ were the first to predict that a particle of ferromagnetic material, below a critical particle size, would consist of a single magnetic domain. This critical particle size has been estimated by Kittel² and others, and an approximate figure for a spherical sample of the common ferromagnetic materials is a radius of 150 Å. More recently, Kondorsky,³ Brown,⁴ and Frei, Shtrikman, and Treves⁵ have treated this subject more rigorously, and have discussed the several different meanings the term "single domain particle" may have. Except where specifically stated otherwise, in this article "single domain particle" will mean a particle that is in a state of uniform magnetization at any field.

Let us consider such a particle whose total magnetic moment μ is directed at an angle θ to an applied field H , as shown in Fig. 1(a). If there are no additional anisotropic terms in the energy, the energy of this particle can be given as $-\mu H \cos\theta$. If we have an assembly of such particles at temperature T that in some manner has achieved a state of thermodynamic equilibrium with the field, there will be a Boltzmann distribution of θ 's over the particle assembly. The fraction of the total magnetization that has been aligned by the field

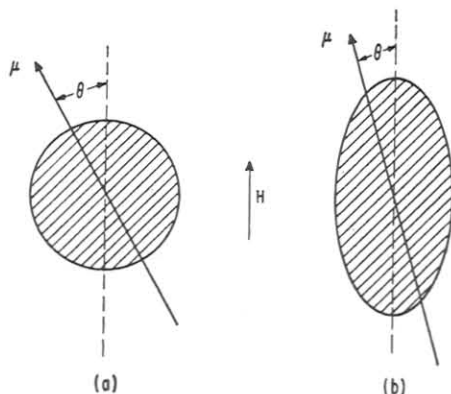


FIG. 1. (a) Isotropic single domain particle, (b) single domain particle with uniaxial anisotropy.

¹ J. Frenkel and J. Dorfman, *Nature* **126**, 274 (1930).

² C. Kittel, *Phys. Rev.* **70**, 965 (1946).

³ E. Kondorsky, *Doklady Akad. Nauk S.S.S.R.* **82**, 365 (1952).

⁴ W. F. Brown, Jr., *J. Appl. Phys.* **29**, 470 (1958).

⁵ Frei, Shtrikman, and Treves, *Phys. Rev.* **106**, 446 (1957).

is calculated by averaging $\cos\theta$ over the Boltzmann distribution, which yields the familiar Langevin function, $L(x) = \coth x - 1/x$, where $x = \mu H/kT$. For low fields this function is approximately $\mu H/3kT$. For high fields, it gives $1 - kT/\mu H$ for the form of the approach to saturation.⁶

This treatment of the thermal equilibrium magnetization properties of an assembly of isotropic single domain particles is, of course, analogous to the Langevin treatment of atomic paramagnetism. It differs only in that the moment μ we are dealing with is not that of a single atom, but rather of a single domain ferromagnetic particle, which may contain more than 10^5 atoms ferromagnetically coupled by exchange forces.

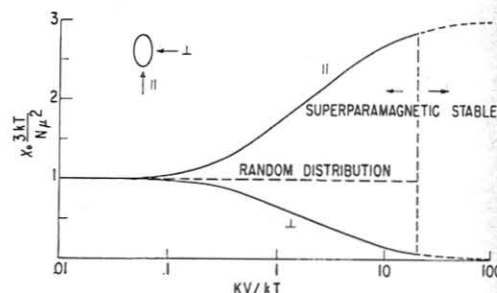


FIG. 2. Dependence on KV/kT of the initial susceptibility, χ_0 , measured parallel and perpendicular to the axis of a uniaxially anisotropic particle. N is the number of particles per unit volume.

In nature, single domain particles are not completely isotropic in their properties, but will have anisotropic contributions to their total energy associated with the external shape of the particle, imposed stresses, or the crystal structure itself. Consider first an anisotropy that is uniaxial in symmetry, in particular, one of the simplest form, $E_K = KV \sin^2\theta$. In this expression, θ is the angle between the moment and the symmetry axis of the particle, V is the volume of the particle, and K is the anisotropy energy per unit volume. If a field H is applied along the symmetry axis, as in Fig. 1(b), the energy of the particle shown will be $KV \sin^2\theta - \mu H \cos\theta$. We will now have a different Boltzmann distribution of θ 's in thermal equilibrium than we had without the anisotropy term, and the magnetization curve will no

⁶ E. C. Stoner, *Phil. Trans. Roy. Soc. (London)* **A235**, 165 (1936).

longer be a simple Langevin function. The low field approximation for the fractional magnetization will depend on the value of KV/kT , and for $K > 0$ will vary from $\mu H/3kT$ for $KV \ll kT$ to $\mu H/kT$ for $KV \gg kT$. This is shown by the upper curve in Fig. 2. If the field applied is perpendicular to the easy symmetry axis, the susceptibility will decrease with increasing KV/kT , as the lower curve in Fig. 2 demonstrates. However, if one has an assembly of uniaxially anisotropic particles with the directions of their symmetry axes distributed at random, the initial fractional magnetization of the random assembly remains at $\mu H/3kT$ for all values of KV/kT .

The dominant term governing the approach to saturation of such particles is found to be still $kT/\mu H$, regardless of the direction of H and the magnitude of KV/kT .

If instead of a uniaxial anisotropy the single domain particles had an anisotropy of cubic symmetry, the $\mu H/3kT$ expression for the initial fractional magnetization and the $1 - kT/\mu H$ expression for the approach to saturation are both applicable for any direction of the applied field and any value of KV/kT .

APPROACH TO THERMAL EQUILIBRIUM

We have until now discussed the magnetization properties to expect of an assembly of identical single domain particles in thermal equilibrium with the applied field. The familiar permanent magnet properties of stable single domain particles are, of course, strikingly different, and are associated with particles that are not in a state of thermodynamic equilibrium. Let us now consider under what conditions an assembly of single domain particles can achieve thermal equilibrium in a time short compared with the time of an experiment.

One means of approaching equilibrium is by physical rotation of the individual particles, which is possible if the particles are suspended in a liquid medium. Elmore⁷ has found that a colloidal suspension of single domain magnetic iron oxide particles shows paramagnetic behavior of the type we have been discussing. With this mechanism the factor determining the rate of approach to equilibrium is the viscosity of the medium in which the particles are suspended.

This problem may be treated in the way Debye⁸ has treated the dielectric properties of solutions containing molecules with permanent electric dipole moments, since in this case it is the viscosity of the solution that opposes the rotation of the elementary dipoles. There are, of course, many other parallels between Langevin paramagnetism and the electric properties of substances containing permanent electric dipole moments. For example, the phenomenon of "hindered rotation" in

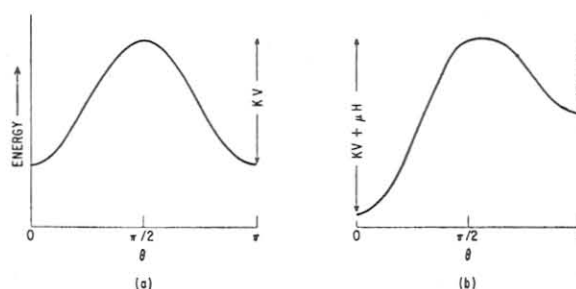


FIG. 3. Dependence of energy of a uniaxially anisotropic particle on the angle between the magnetization and the easy symmetry axis. (a) Without field, (b) with a small field applied along $\theta = 0$.

dielectrics⁹ is caused by the analogy of anisotropy in ferromagnets.

In a solid, the above relaxation mechanism will, of course, not be available. In 1949, Néel¹⁰ pointed out that if a single domain particle were small enough, thermal fluctuations could cause its direction of magnetization to undergo a sort of Brownian rotation. He also derived the conditions under which an assembly of such particles could come to thermal equilibrium in a given time.

Consider an assembly of aligned uniaxial particles such as the one in Fig. 1(b) that are first fully magnetized along the easy symmetry axis. After the field is removed, the resulting remanence will vanish as $M_r = M_s \exp(-t/\tau)$, where M_s is the full magnetization, t is the time after removal of the field, and τ is the relaxation time for the process. The relaxation time is given by $1/\tau = f_0 \exp(-KV/kT)$, where f_0 is a frequency factor of the order of 10^9 sec^{-1} . The source of the exponential is clear from Fig. 3(a), which shows the dependence of energy on θ for this particle in zero field. To approach the zero remanence corresponding to thermal equilibrium, a sufficient number of particles must be reversed by thermal activation over the energy barrier KV , the probability of which process is proportional to $\exp(-KV/kT)$.

For particles with an anisotropy of cubic symmetry, the energy barrier between adjacent easy directions will appear in the exponential. Using the standard definition of the first order cubic anisotropy constant, the barrier is $KV/4$ for $K > 0$ ([100] easy direction) and $KV/12$ for $K < 0$ ([111] easy direction).

Because of the exponential dependence of τ on particle volume, there is a fairly well defined particle size at which the transition to stable behavior occurs. For a spherical iron particle whose only source of anisotropy is its first order crystal anisotropy, a particle 115 Å in radius will have a relaxation time of 10^{-1} sec at room temperature and hence will reach thermal equilibrium almost instantaneously. But a particle 150 Å in radius will have a relaxation time of 10^9 sec and hence

⁷ W. C. Elmore, Phys. Rev. **54**, 1092 (1938).

⁸ P. Debye, *Polar Molecules* (Chemical Catalog Company, New York, 1929).

⁹ J. H. Van Vleck, Ann. N. Y. Acad. Sci. **40**, 293 (1940).

¹⁰ L. Néel, Compt. rend. **228**, 664 (1949); Ann. géophys. **5**, 99 (1949).

will be extremely stable. There will in general be only a narrow range of particle sizes in which the relaxation times will be of the order of experimental times, for which measurable "magnetic viscosity" effects would be expected.

As a rough measure of the size for transition to stable behavior, one can take the size corresponding to $\tau = 10^2$ sec. This occurs when the energy barrier is equal to approximately $25 kT$. For a given particle, the temperature at which this occurs has been called the "blocking temperature." We have calculated the particle radius for which room temperature is the "blocking temperature" for various materials, considering only first-order anisotropy constants. We obtain 40 Å for h.c.p. cobalt, 125 Å for iron, and 140 Å for f.c.c. cobalt. For an elongated iron particle with the optimum shape anisotropy, the calculated critical volume is equal to that of a sphere of 30 Å radius.

It should be pointed out that the energy barriers calculated here have been based on the assumption that the reversal process is one of coherent rotation. However, at least where the source of the anisotropy is the particle shape, it is possible that there exists a noncoherent reversal process with a lower energy barrier, such as the "magnetization curling" process described by Frei, Shtrikman, and Treves.⁸ Néel avoids this difficulty by putting the coercive force of the particle (when it is stable) in the exponential, which then corresponds to whatever reversal process the particle finds easiest.

Experimental evidence that for an elongated particle the energy barrier for coherent rotation is larger than that determining the transition to stable behavior appears in Fig. 4 of a paper by Livingston and Becker.¹¹ Quantitative comparison of the critical particle sizes for the appearance of coercive force for cobalt particles

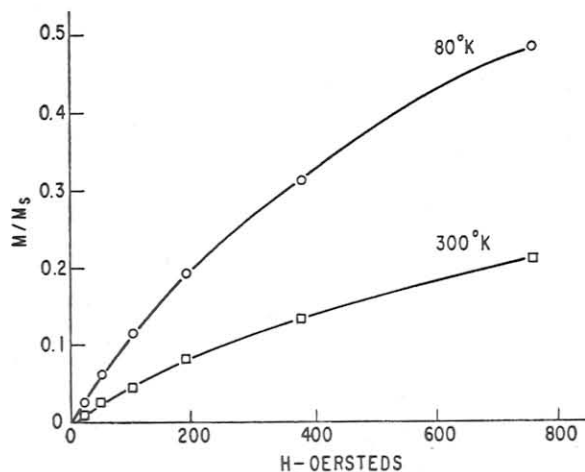


FIG. 4. Magnetization curves of 25 Å radius nickel catalyst particles showing superparamagnetic behavior (after Heukelom, Broeder, and Van Reijen).

¹¹ J. D. Livingston and J. J. Becker, *Trans. Am. Inst. Mining, Met., Petrol. Engrs.* **212**, 316 (1958).

before and after elongation by plastic deformation indicates that the energy barrier for the elongated particles is more nearly equal to $KV/2$ than to KV .

The previous discussion of relaxation times has been for the case of zero applied field. We now consider the case of a reversing field H applied to an aligned assembly of uniaxial particles that is stable in zero field. The energy barrier for rotation is now less than KV , approximately $KV - \mu H$ for small H , as shown in Fig. 3(b). More accurately, the barrier is $KV - \mu H + \mu^2 H^2 / 4KV$. When H is large enough that this energy barrier is decreased to approximately $25kT$, the reversal process can be thermally activated within the time of experiment. (In this treatment we neglect changes in the pre-exponential factor, f_0 , with H .) This criterion yields* $2KV[1 - 5(kT/KV)^{1/2}]/\mu$ as the coercive force of these particles. This formula describes the coercive force from $T=0$, where it yields the Stoner-Wohlfarth¹² value, $2KV/\mu$, to $T=KV/25k$, where the coercive force becomes zero and ideal superparamagnetism sets in.

EXPERIMENTAL OBSERVATION OF SUPERPARAMAGNETISM

We have seen that the magnetization behavior of single domain particles in thermodynamic equilibrium at all fields is identical with that of atomic paramagnetism except that an extremely large moment is involved, and thus large susceptibilities are involved. Because of these similarities and differences, such thermal equilibrium behavior has been termed "superparamagnetism."¹³ This behavior has also been discussed in the literature under several other names, including "apparent paramagnetism,"¹⁴ "collective paramagnetism,"¹⁵ "quasiparamagnetism"¹⁶ and "subdomain behavior."¹⁷

An operational definition of superparamagnetism would include at least two requirements. First, the magnetization curve must show no hysteresis, since that is not a thermal equilibrium property. Second, except for particle interaction effects which will be discussed later, the magnetization curve for an isotropic sample must be temperature dependent to the extent that curves taken at different temperatures must approximately superimpose when plotted against H/T after correction for the temperature dependence of the spontaneous magnetization.

Perhaps the first example in the literature of clearly superparamagnetic behavior under the above definitions is the work of Heukelom, Broeder, and Van

* An earlier attempt, as set down in reference 24, by one of us (C.P.B.) to obtain this expression is totally wrong. We are indebted to Dr. E. P. Wohlfarth for pointing out this error.

¹² E. C. Stoner and E. P. Wohlfarth, *Phil. Trans. Roy. Soc. (London)* **A240**, 599 (1948).

¹³ C. P. Bean, *J. Appl. Phys.* **26**, 1381 (1955).

¹⁴ Weil, Gruner, and Deschamps, *Compt. rend.* **244**, 2143 (1957).

¹⁵ A. Knappwost, *Z. Elektrochem.* **61**, 1328 (1957).

¹⁶ R. Hahn and E. Kneller, *Z. Metallk.* **49**, 426 (1958).

¹⁷ A. E. Berkowitz and P. J. Flanders, *J. Appl. Phys.* **30**, 111S (1959), this issue.

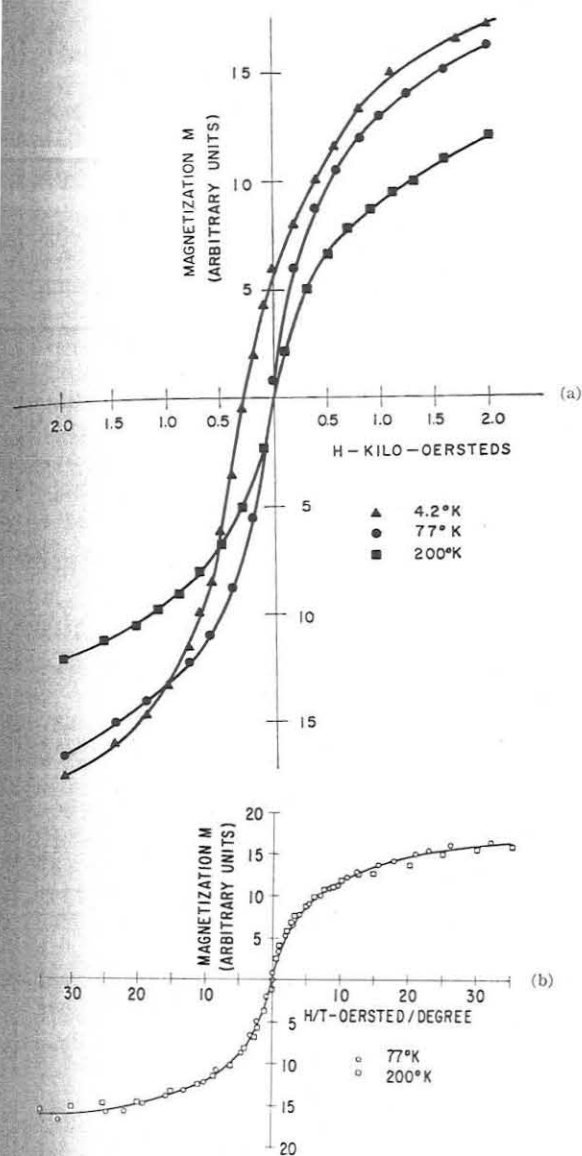


Fig. 5. (a) Magnetization curves of 22 Å radius iron particles in mercury. (b) H/T superposition of these data (after Bean and Jacobs).

Reijer¹⁸ on finely divided silica-supported nickel catalysts. Figure 4 shows magnetization curves taken from this paper, and it is clear that rough H/T superposition is present.

More recently Bean and Jacobs¹⁹ have demonstrated superparamagnetic behavior of fine particles of iron suspended in mercury. Figure 5(a) shows their magnetization curves at various temperatures, and Fig. 5(b) shows the approximate H/T superposition observed for their 77°K and 200°K data. Although these particles are therefore superparamagnetic at these temperatures,

it is clear from Fig. 5(a) that at 4.2°K some of them are stable.

Figure 6 shows the excellent H/T superposition obtained by Becker²⁰ with small particles of cobalt precipitated out of solid solution in a copper-2% cobalt alloy. More recently, fine particles of iron precipitated out of solid solution in β -brass¹⁷ have been shown to fulfill the requirements for ideal superparamagnetic behavior. Some degree of superparamagnetic behavior has also been reported for micropowders of nickel and cobalt ferrites,²¹ fine particles of γ - Fe_2O_3 ,¹⁵ and small regions of ordered Ni_3Mn in a disordered nonmagnetic Ni-Mn matrix,¹⁶ but in these cases full H/T superposition has not yet been observed.

APPLICATIONS

Magnetic measurements on superparamagnetic particles have already found application as a means of determining particle size. We have seen that the magnetization curve for an isotropic assembly of paramagnetic particles of identical moment μ is the Langevin function with $\mu H/kT$ as its argument. Thus by fitting a curve measured at a given temperature to the Langevin formula we should be able to determine μ . If we know the spontaneous magnetization per unit volume of the particle, I_s , the particle volume is given by $V = \mu/I_s$.

However, two problems immediately arise. Firstly, there may be some doubt as to whether the value of I_s appropriate to bulk material may be applied to such small particles. This will be discussed later. Secondly, it is an experimental fact that as yet no magnetization curves for superparamagnetic particles have been reported which can be fit to a simple Langevin function. It is usually presumed that this is caused by the presence of a distribution of particle sizes, and hence a distribution of particle moments.

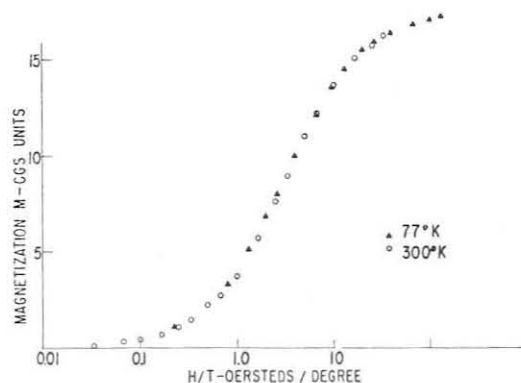


Fig. 6. H/T superposition demonstrating superparamagnetic behavior of 27 Å radius cobalt particles formed by precipitation in a copper-2% cobalt alloy (after Becker).

¹⁸ Heukelom, Broeder, and Van Reijen, *J. chim. phys.* **51**, 474 (1954).

¹⁹ C. P. Bean and I. S. Jacobs, *J. Appl. Phys.* **27**, 1448 (1956).

²⁰ J. J. Becker, *Trans. Am. Inst. Mining, Met., Petrol. Engrs.* **209**, 59 (1957).

²¹ A. E. Berkowitz and W. J. Schuele, *J. Appl. Phys.* **30**, 134S (1959), this issue.

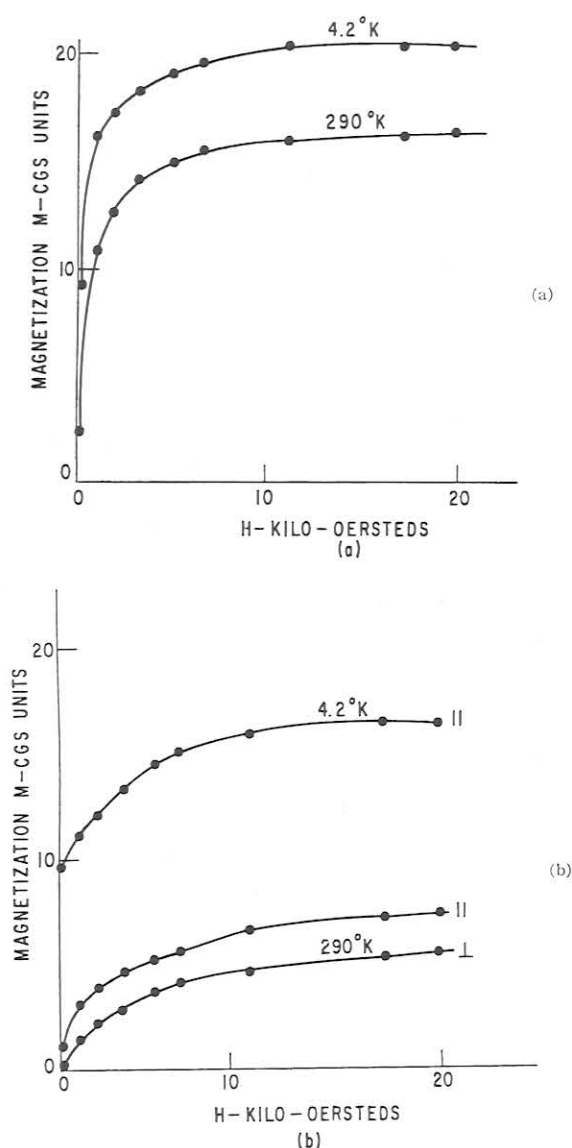


FIG. 7. Magnetic evidence for the subdivision of precipitated cobalt particles during heavy deformation of a copper-2% cobalt alloy. (a) Before deformation. (b) After drawing from ingot 1 in. square to wire 0.012 in. in diameter. For 290°K, data are shown as taken both perpendicular and parallel to the direction of deformation (after Weil).

If one has a distribution in particle sizes, various parts of the magnetization curve can be used to obtain various averages over the particle size distribution.^{20,22} The initial susceptibility is sensitive to the larger particles present, whereas the approach to saturation is largely governed by the smaller particles. Measurement to both averages yields information on the width of the size distribution.

Elmore⁷ and Henning and Vogt²³ have assumed a Gaussian distribution in sizes, and have determined

²² J. W. Cahn, Trans. Am. Inst. Mining, Met., Petrol. Engrs. 209, 1309 (1957).

²³ W. Henning and E. Vogt, Z. Naturforsch. 12a, 754 (1957).

the parameters of the distribution by fit to the experimental curve. Becker²⁰ and Hahn and Kneller¹⁶ have employed a very different distribution function. The fit to the experimental curve is not sensitive to the details of the distribution.

Determination of particle size employing superparamagnetism has been used to study the kinetics of growth of precipitated particles of cobalt in copper-cobalt alloys. Saturation magnetization measurements show that the precipitation is complete within the first few minutes at elevated temperatures, and that the subsequent process is resolution of the smaller particles and growth of the larger particles.^{20,24} It has recently been shown²⁵ that the coarsening kinetics fit the form predicted by Greenwood²⁶ for such a process. These magnetic measurements have also been used to correlate precipitate particle size with the mechanical properties of these alloys.^{11,25} It has been found that for alloys containing various percentages of cobalt aged at various temperatures the peak yield stress always occurs at approximately the same particle size.

Weil²⁷ has employed the size-determining feature of superparamagnetic behavior to demonstrate the decrease in size of precipitate particles in a copper-cobalt alloy caused by heavy deformation. Figure 7 shows the magnetization curves of a sample before and after a heavy swaging and drawing operation. It can be seen that the initial susceptibility at room temperature, and thus the average particle size, has been decreased by the deformation. It is also seen that the deformed sample is anisotropic in its magnetic properties. It has been shown that this anisotropy is a result of the change of shape of the particles during deformation.¹¹ This latter study included data for much smaller deformations, where change in particle size is apparently minor and the shape change has the major effect on the magnetic properties.

Determination of particle size from magnetization curves was applied by Heukelom *et al.*¹⁸ to the study of the formation of nickel particles for use as catalysts. Dietz and Selwood²⁸ are currently using these techniques in studying the effects of adsorbed gases on nickel catalysts.

The above applications have been based on the thermal equilibrium behavior of superparamagnetic particles. Important application has also been made of the transition from superparamagnetic to stable single domain behavior, as described by Néel.

If one has an assembly of exclusively single domain particles, the remanence observed at a given temperature should be a measure of the amount of material

²⁴ J. J. Becker, Trans. Am. Inst. Mining Met. Petrol. Engrs. 212, 138 (1958).

²⁵ J. D. Livingston (submitted to Trans. Am. Inst. Mining, Met. Petrol. Engrs.).

²⁶ G. W. Greenwood, Acta Met. 4, 243 (1956).

²⁷ L. Weil, J. phys. radium (to be published).

²⁸ R. E. Dietz and P. W. Selwood, J. Appl. Phys. 30, 101S (1959), this issue.

with particle volume greater than that just stable at this temperature. Thus by following the increase of remanence with decreasing temperature, one can find how much material lies in various ranges of volume. In principle, one could then determine the entire distribution of particle volumes. Weil has applied this technique both to nickel powders²⁹ and to cobalt particles in a copper-cobalt alloy.³⁰ Figure 8(a) shows his data on coercive force and ratio of remanence to saturation magnetization for a nickel powder as a function of temperature. Figure 8(b) shows the distribution in particle volumes inferred from these data. His calculations assume that the dominant anisotropy of the particles is uniaxial, so that the remanence to be expected of a randomly oriented set of completely stable particles is one-half the saturation magnetization. Should the particle anisotropies be cubic in symmetry, as is believed to be the case for cobalt particles in copper-cobalt alloys, the interpretation should be modified both for the different conditions for superparamagnetic behavior and for a different remanence ratio. Henning and Vogt²³ have recently applied Weil's technique to determine particle size distribution for cobalt and iron particles in mercury and found the distribution so derived to be consistent with the superparamagnetic magnetization curve at an elevated temperature.

The field which first led Néel to the concept of thermal fluctuations in single domain particles was that of paleomagnetism. Many rocks contain small amounts of magnetic material, largely mixed titanium-iron oxides. Much work has been directed towards measuring the intensity and direction of remanent magnetization in various kinds of rocks, with the thought that such data yield information on the strength and direction of the local earth's magnetic field at the time the rock was formed. Blackett³¹ says, "The detailed study of the natural magnetization of rocks is likely to allow us to trace back to the beginning of geological time both the history of the earth's magnetic field and the motion of continental masses relative to each other and to the geographical pole." The ability of the magnetic rocks to "remember" an earlier magnetic field depends on their ability to exist in thermal equilibrium with the earth's field at a stage early in their formation but to be later "frozen" in a state of magnetization stable against later changes of the direction and strength of the field.

The original establishment of thermal equilibrium may occur by either of the relaxation mechanisms we discussed earlier. In sedimentary rocks the small grains of magnetic material can align themselves with the direction of the earth's field by physical rotation of the particles while the sediment is still wet and uncom-

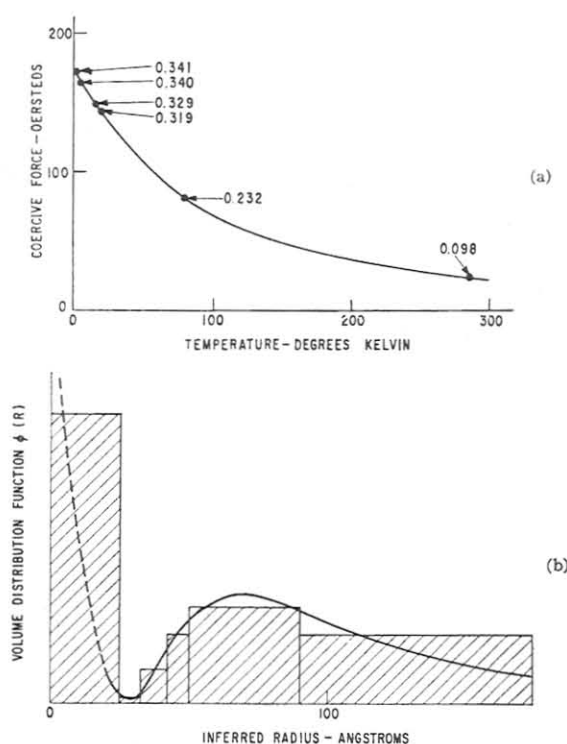


FIG. 8. (a) Coercive force vs temperature for a nickel powder. Numbers along the curve indicate the ratio of remanence to saturation magnetization at the various temperatures. (b) Particle size distribution inferred from these data (after Weil).

pressed. Later, when the sediment becomes relatively hard rock, this relaxation mechanism is lost and the magnetization is stable.

In igneous rocks, on the other hand, when the material becomes magnetic by cooling through its Curie point the rock is already solid and physical rotation of the particles cannot occur. At high temperatures, however, energy barriers can be quite small and kT can be large. Single domain particles may therefore be able to reach thermal equilibrium by Néel's relaxation mechanism, but will become stable when cooled through their "blocking temperature." Their state of magnetization may then be stable against later changes of the earth's field, and will maintain a memory of the strength and direction of the field present when the particles cooled through their blocking temperature. Calculations of blocking temperatures will be slightly different from those discussed earlier, since now "experimental times" are not of the order of 10^2 sec, but are geologic times.

That the original direction of magnetization of rocks can often be extremely stable has been demonstrated by Graham.³² Figure 9(a) shows a rock stratum as initially magnetized. Graham has studied strata that were deformed at remote times and has found that in many cases the initial direction of magnetization has remained, as shown in Fig. 9(b).

³² J. W. Graham, *J. Geophys. Research* 54, 131 (1949).

²⁹ L. Weil, *J. Chim. Phys.* 51, 715 (1954).

³⁰ L. Weil and L. Gruner, *Compt. rend.* 243, 1629 (1956).

³¹ P. M. S. Blackett, *Lectures on Rock Magnetism* (Weizmann Science Press, Jerusalem, 1956).

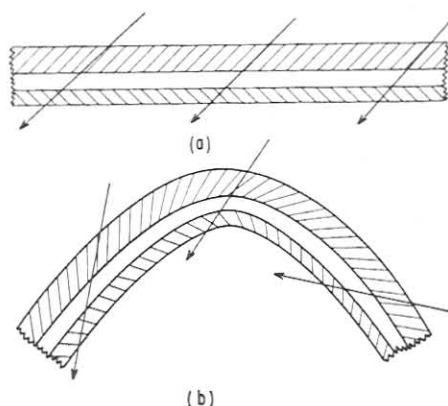


FIG. 9. Indication of the degree of stability of the direction of magnetization shown by some rocks. (a) Rock stratum as initially magnetized. (b) Deformed stratum, still maintaining its original direction of magnetization (after Graham).

The magnetic memory effects of rocks may be alternatively explained by thermal effects on the magnetic properties of multidomain particles. Further experiments are necessary to establish the mechanisms of blocking for individual cases.

SPONTANEOUS MAGNETIZATION

One interesting problem is that of determining whether the spontaneous magnetization, extrapolated to absolute zero of temperature, depends upon particle size. To take a specific instance, consider a small particle of cobalt imbedded in copper. Since there is a contact potential between these two metals, one expects an electronic redistribution at their interface and associated with this electron transfer a possible alteration in the spontaneous magnetization ascribed to the cobalt. This alteration would be a function of the particle size. It is interesting to note that experiments on this system show the saturation magnetization to be identical, within 2%, for radii from 21 Å to 77 Å.^{20,33} In the case of the smaller particles 25% of the atoms are surface atoms. Similar results have been obtained by Luborsky³⁴ on small particles of iron suspended in mercury. On the other hand, evidence has been presented by Selwood and his students to support the thesis that adsorption of hydrogen on nickel particles decreases the spontaneous magnetization by chemical bonding between the hydrogen and the surface nickel atoms.²⁸ These experiments are not in essential disagreement with one another, since the electronic redistribution at the surface would be expected to be much more pronounced for contact with vacuum or chemisorbed hydrogen than for contact with a similar metal. It is clear that further studies of this effect, in addition to aiding the understanding of catalysis, may cast light upon the nature of metallic surfaces and interfaces.

³³ Bean, Livingston, and Rodbell, *J. phys. radium* (to be published, 1959).

³⁴ F. E. Luborsky, *J. Appl. Phys.* **29**, 309 (1958).

A second problem is the measure of the spontaneous magnetization at elevated temperatures. Does the presence of the surface atoms with their deficiency of magnetic neighbors cause the spontaneous magnetization of a fine particle to fall off more drastically with temperature than occurs for bulk materials? Much of the expectations on this subject have been derived from the experimental and theoretical work on thin films. Bloch deduced that a two-dimensional ferromagnet, considering exchange forces alone, would not have a second order phase transition (Curie temperature). Klein and Smith³⁵ extended this work to the consideration of films with a finite thickness and traced the transition from the case of the bulk material to that of an infinitely thin sheet. The experimental work of Crittenden and Hoffman³⁶ on nickel films is in qualitative agreement with this theory and indicates, for instance, that for thicknesses less than 20 Å, nickel films have no spontaneous magnetization at room temperature. These experimental and theoretical results cannot be applied to fine particles because the behavior deduced by Bloch is critically dependent on the fact that the thin film is infinite in two directions.

In the case of fine particles it does not seem immediately apparent that the spontaneous magnetization, as distinguished from the saturation magnetization, can be measured, since to line up the moments of a superparamagnetic assembly may require such fields as to alter the moment of each particle. Fortunately, if one examines the formula for the initial susceptibility, $\chi = NVI_s^2/3kT$, one sees that it is proportional to the square of the spontaneous magnetization. This measurement may be made in arbitrarily low fields to insure that the moment of each particle is unaffected by the field. Thus the variation of χT with temperature will measure the variation of I_s^2 with temperature. An abnormally fast variation of I_s with temperature, as suggested by the work on thin films, would become immediately apparent.

Knappwost and Illenberger³⁷ have very recently employed this technique. They find, in contrast to earlier interpretations,³⁸ that the spontaneous magnetization of cobalt particles 7 Å in radius precipitated from copper is independent of temperature up to 300°C. Using similar reasoning to that presented above, Kneller^{38a} has deduced the course of spontaneous magnetization with temperature for 12.5 Å particles of ordered Ni₃Mn precipitated from the disordered nonmagnetic material. He finds the Curie temperature of the ordered material to be 270°K for the small particles as against 740°K for the bulk material. It is not completely convincing to ascribe this difference to particle size inasmuch as

³⁵ M. J. Klein and R. S. Smith, *Phys. Rev.* **81**, 378 (1951).

³⁶ E. C. Crittenden, Jr., and R. W. Hoffman, *Revs. Modern Phys.* **25**, 310 (1953).

³⁷ A. Knappwost and A. Illenberger, *Naturwissenschaften* **45**, 238 (1958).

³⁸ A. Knappwost, *Z. physik. Chem. (Frankfurt)* **12**, 30 (1957).

^{38a} E. Kneller, *Z. Physik* **152**, 574 (1958).

one may question whether this nonequilibrium alloy has completely ordered particles of Ni_3Mn .

Recently, J. W. Cahn has succeeded in producing extremely small particles of cobalt precipitated from a $\text{Cu}-2\%$ Co solid solution by quenching from 1000°C directly to the annealing temperature. The magnetization curves shown in Fig. 10(a) were taken by I. S. Jacobs and P. E. Lawrence on a sample Cahn had annealed for 16 hr at 400°C . The average radius calculated from the initial slope of the magnetization corresponds to a radius of 8 Å or 200 atoms per particle. One notices that only a minute fraction of these particles becomes stable at 4.2°K . This is quite reasonable since $[K_1V/kT] \sim 4$ for this temperature—a value significantly lower than the critical value, 300, for negative cubic K_1 indicated earlier. The constancy of spontaneous magnetization with temperature is tested by plotting the magnetization as a function of H/T , as in Fig. 10(b). It is seen that, in the region of initial magnetization, the curves substantially fall upon one another. In this figure, curve A represents the Langevin function that would be followed if the particles were uniform in size and if $KV/kT \ll 1$. Curve B is the experimental curve for elevated temperatures whose deviation from A we ascribe to a particle size distribution. Using the same deviation from a uniform distribution, we have calculated curve C on the assumption that $KV/kT \gg 1$. This calculation assumes that the magnetization is constrained to lie in the $[111]$ directions and is analogous to an earlier calculation on extreme uniaxial anisotropy.¹³ In summary, the experiment of Cahn, Jacobs, and Lawrence appears to show that the spontaneous magnetization of these particles is independent (within 4%) of temperature from 4.2°K to 300°K . In the next section we shall show that this conclusion is not as surprising as it may seem at first sight.

ATOMIC VIEWPOINT

It is illuminating to derive the magnetization of very small clusters, containing two or three atoms. The problem is to deduce the energy levels that result from the interaction of the atoms with one another and with the external field. Having these levels, a simple application of statistical mechanics yields the magnetization. We shall consider assemblies of one, two, and three spin $\frac{1}{2}$ particles linked by an exchange interaction. The Hamiltonian that expresses the couplings may be written as

$$\mathcal{H} = -J \sum_{i,j} \mathbf{S}_i \cdot \mathbf{S}_j - g\mu_B \sum_i \mathbf{S}_i \cdot \mathbf{H} = \mathcal{H}_J + \mathcal{H}_H,$$

where J is the exchange integral, $\mathbf{S}_i$ and $\mathbf{S}_j$ the spin operators (in units of $\hbar$) that act upon the i th and j th atom, respectively, g is the spectroscopic splitting factor (closely equal to 2), μ_B is the Bohr magneton, and H is the external field. The first term in the Hamiltonian is the exchange interaction while the second is

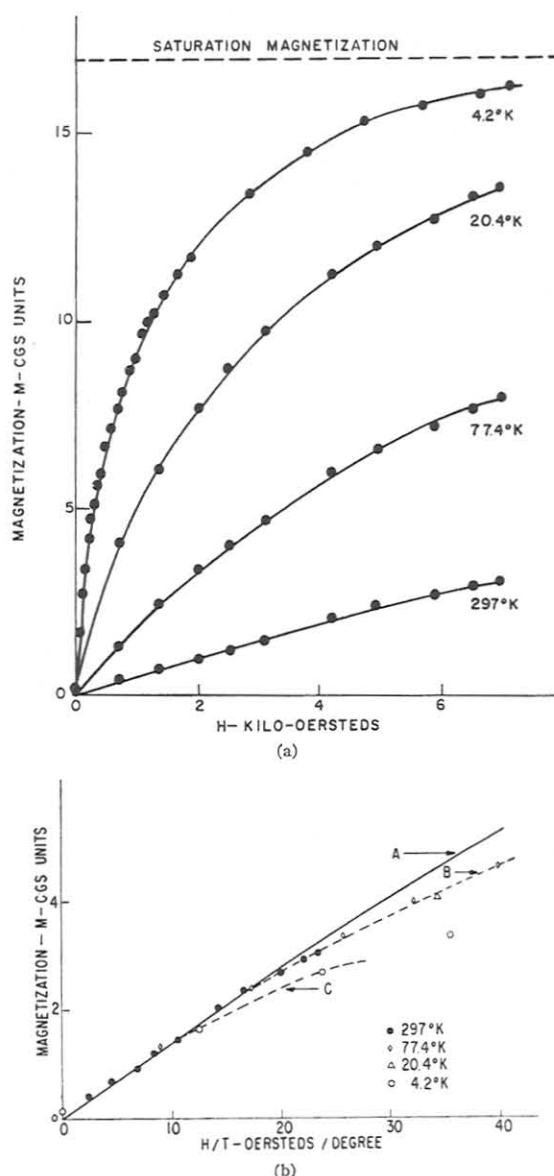


FIG. 10. (a) Magnetization curves for a copper-2% cobalt alloy containing precipitate particles approximately 8 Å in radius. (b) H/T superposition of low field data (after Cahn, Jacobs, and Lawrence).

the interaction with the external field. We have not included the magnetostatic dipole-dipole interactions, crystal field interactions, or anisotropic exchange that give rise to anisotropy since these are not essential to our result. If we pick as trial functions the linear combinations of spin parallel and antiparallel to H illustrated in Fig. 11, we find that these are indeed eigenfunctions with the relative eigenvalues, $\langle n | \mathcal{H} | n \rangle$, noted in the figure.

Each subset of levels, corresponding to a given exchange energy, has the normal Brillouin susceptibility

$$\chi = NS(S+1)g^2\mu_B^2/3kT,$$

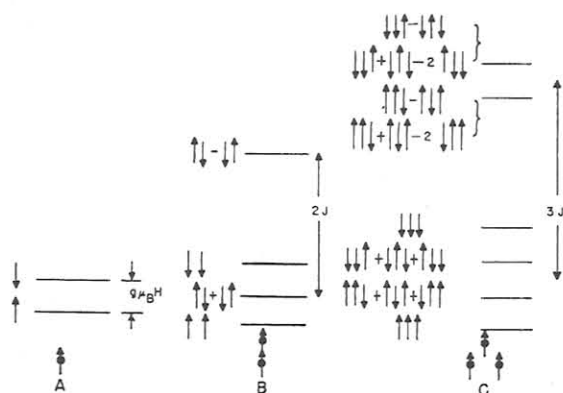


FIG. 11. Energy levels in a magnetic field H of a single spin, two spins with an exchange coupling J , and three spins with an exchange coupling J .

where $2S+1$ is the multiplicity of the given subset of levels. [In Fig. 11(c), the upper levels are two degenerate subsets, each of spin $\frac{1}{2}$.] The susceptibility, therefore, considering all levels is

$$\chi = \frac{Ng^2\mu_B^2 \sum_n S(S+1) \exp(-\langle n | \mathcal{H}_J | n \rangle / kT)}{3kT \sum_n \exp(-\langle n | \mathcal{H}_J | n \rangle / kT)}.$$

In the spirit of the previous sections, the quotient of sums on the right-hand side of the expression above is proportional to the square of the spontaneous magnetization. A single spin, viewed in this manner, has a spontaneous magnetization that is independent of temperature. Two coupled spins have the temperature dependence indicated in Fig. 12. For three spins, each a nearest neighbor to the others, the temperature dependence is less drastic than for two because of the extra interactions. For comparison, we have plotted the square of the magnetization deduced from the Bloch $T^{\frac{1}{2}}$ law for a f.c.c. material.³⁹ We notice that the spontaneous magnetization falls off more rapidly with temperature in this case than for any of the small aggregates considered. The physical reason for this behavior lies in the fact that the excited states (spin waves) of the large specimen are very close to the ground state and hence may be excited thermally more easily than the widely spaced states associated with, say, three spins.

ADDITIONAL QUESTIONS

In the preceding sections, we have outlined the theoretical and experimental foundations of superparamagnetism. There remain, however, many interesting questions. One of these is that of the determination of the pre-exponential frequency factor in the expression for the relaxation time. We shall not treat this question, inasmuch as Brown⁴⁰ reviews and extends the theory of this quantity in another conference paper.

³⁹ F. J. Dyson, Phys. Rev. **102**, 1230 (1956).

⁴⁰ W. F. Brown, Jr., J. Appl. Phys. **30**, 130S (1959), this issue.

We note, however, that the possibility of tunneling should be investigated theoretically and experimentally. By tunneling we mean the possibility of transitions at absolute zero from a state of complete alignment towards a state of zero over-all magnetization. If this process were to exist, it could help explain the puzzling behavior of Weil's nickel particles shown in Fig. 8. In this experiment, apparently at least a quarter of the particles did not reach their blocking temperature at 4.2°K. Further, the remanence does not increase appreciably on lowering the temperature to 2°K, implying a rather strange gap in the particle distribution in Weil's explanation.

Another effect that may account for Weil's result is that of clumping of the particles. Figure 13 shows an electron micrograph of rather large particles of iron that have been removed from the mercury in which they were grown. We focus attention on several clumps of roughly spherical particles and have drawn approximate ground states under their mutual magnetic interactions. These interactions cause the magnetization in each clump to assume the more or less complete flux closure indicated. Consequently, they will not contribute to the total remanence of the specimen. Weil⁴¹ and Mayer and Vogt⁴² have called attention to the possibility of these "sponges" in iron amalgams and sintered powders. It is interesting to note that the situation with regard to agglomeration is quite different for particles precipitated in a solid matrix from that for either the amalgams or loose powders. In the latter cases, magnetic attraction causes the particles to agglomerate. In the case of precipitation in a solid matrix, each magnetic particle in the initial stages of precipitation is surrounded by a depleted zone of solid solution and hence the particles tend to avoid one another. A qualification to this statement must be entered if precipitation takes place along grain boundaries or other structural defects.

Another feature of interactions is the possibility of

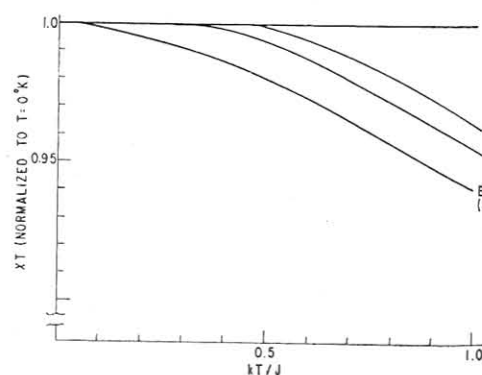


FIG. 12. Dependence of $\chi_0 T$ on temperature for the cases shown in Fig. 11, compared to the temperature decrease of I_s^2 for bulk f.c.c. material based on Bloch's spinwave treatment.

⁴¹ L. Weil, Revs. Modern Phys. **25**, 324 (1953).

⁴² H. Mayer and E. Vogt, Z. Naturforsch. **7a**, 334 (1952).

describing them by an internal field. The first possibility that suggests itself is to employ the Lorentz field.⁴³ In this approximation, the effect of the fields of all the other particles on any particular particle is to cause a net field on this particle in the direction of the average magnetization. This field in a long rod magnetized along its axis is given by

$$H_L = 4\pi M/3,$$

where M is the average magnetization of the specimen. As is well known, the addition of this field to the external field leads to a susceptibility

$$\chi = N\mu^2/3k(T - T_c),$$

where T_c expresses this interaction and has the value

$$T_c = 4\pi N\mu^2/9k.$$

For an alloy containing one volume percent of precipitated cobalt particles 70 Å in radius, this temperature would be approximately room temperature. Experimental results are in striking disagreement with this estimate.²⁰ According to this model, a type of ordered state would occur among the moments at temperatures below T_c . Onsager⁴⁴ pointed out that the Lorentz field is not valid for permanent dipoles, and the results of his approximations are shown in Fig. 14. There is no critical temperature in this theory. We cannot review the recent work in this field⁴⁵ but note that superparamagnetism in precipitation alloys offers an opportunity to critically test these theories.

An interesting type of experiment that, to our knowledge, has not been performed is to measure the superparamagnetism of a system of particles imbedded

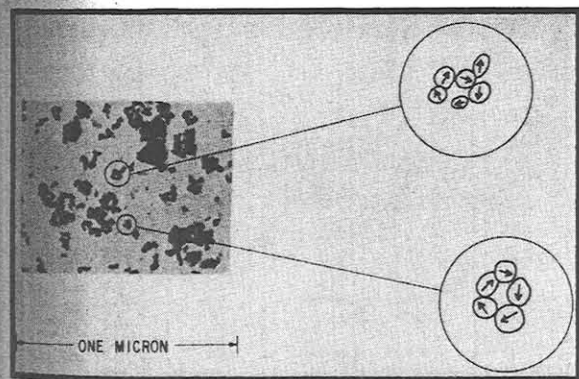


FIG. 13. Electron micrograph showing clustering of small particles of iron. Drawings indicate suggested flux-closure configurations that may keep these clusters from contributing to the remanence of the assembly [after Paine, Mendelsohn, and Luborsky, Phys. Rev. 100, 1055 (1955)].

⁴³ H. A. Lorentz, *The Theory of Electrons* (Dover Publications, New York, 1952), second edition, p. 138.

⁴⁴ L. Onsager, J. Am. Chem. Soc. 58, 1486 (1936).

⁴⁵ e.g., J. W. Smith, *Electric Dipole Moments* (Butterworths Scientific Publications, London, 1955).

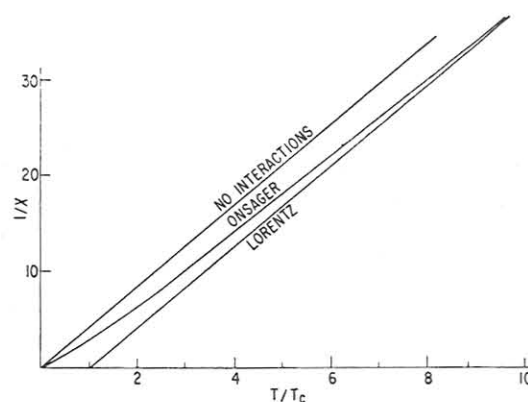


FIG. 14. Effect of particle interactions on the temperature dependence of $1/\chi$, according to Lorentz and according to Onsager. T_c is the critical temperature for the Lorentz interaction and is to be distinguished from a ferromagnetic Curie temperature.

in a magnetic matrix. The effective field on the superparamagnetic particle is the sum of the external field, the cavity field due to the magnetic matrix, and the exchange field caused by the exchange interactions at the particle-matrix interface. This last field, since it is caused by a surface interaction, will be inversely proportional to the radius of the particle and can be interpreted to give a measure of the atomic exchange interaction at the interface. It is appropriate to note that R. Pauthenet⁴⁶ has recently interpreted the paramagnetism of rare earth ions in iron garnets by the atomic counterpart of the effects described above.

Various indirect estimates^{18,19,47} of particle sizes are in accord with the sizes inferred from measurements of superparamagnetism. There have been, however, no direct size measurements of such particles. Advances in electron microscopes and in the techniques of examination of precipitate particles by transmission through thin foils⁴⁸ bring direct measurements in the realm of possibility.

CONCLUSION

In summary, we have seen that the experimental and theoretical foundations of superparamagnetism appear strong enough to support continued application of this technique for the study of particles. In addition, there are opportunities for basic studies on fundamental problems of magnetism utilizing particles in the superparamagnetic size range.

ACKNOWLEDGMENTS

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⁴⁶ R. Pauthenet, J. phys. radium (to be published).

⁴⁷ F. E. Luborsky, J. Phys. Chem. 61, 1336 (1957).

⁴⁸ R. B. Nicholson and J. Nutting, Phil. Mag. 3, 531 (1958).